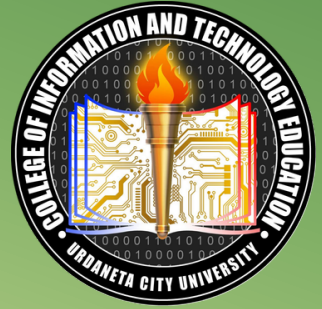
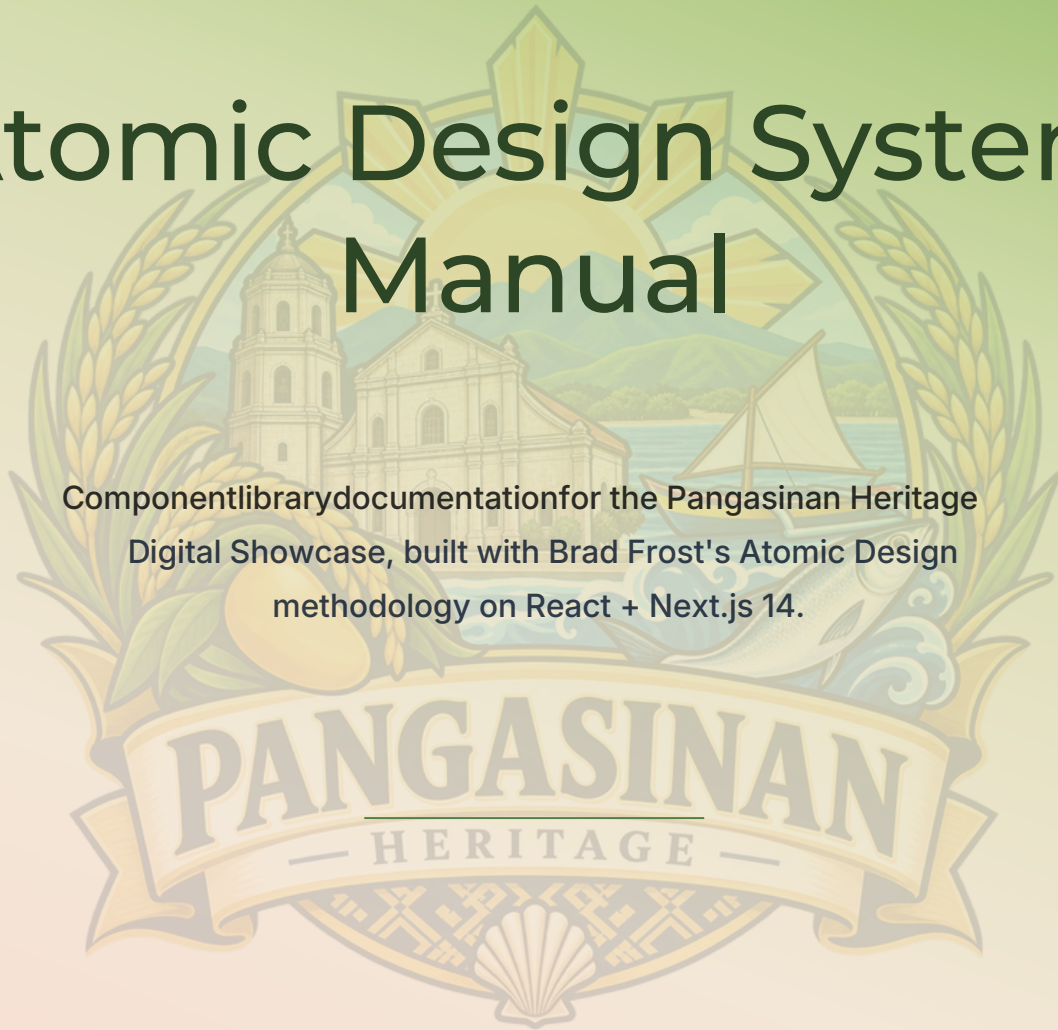




DELIVERABLE 1.2 ACTIVITY 1

Atomic Design System Manual

Component library documentation for the Pangasinan Heritage
Digital Showcase, built with Brad Frost's Atomic Design
methodology on React + Next.js 14.



Elective 4 (Special Topics in IT)

Submitted by: Trishania Marcilla

Instructor: Rey Molano

Overview

This manual documents every component in the Pangasinan Heritage Digital Showcase design system, organized by **Brad Frost's Atomic Design hierarchy**: Atoms → Molecules → Organisms. Each component entry includes: (1) Visual Preview, (2) Usage Context, (3) Responsive Logic, and (4) Code Reference.

Technology Stack: React + Next.js 14 (App Router) · TypeScript · Tailwind CSS v3 · Google Fonts (Inter + Playfair Display) · Inline SVG icons · Static export (JAMstack)

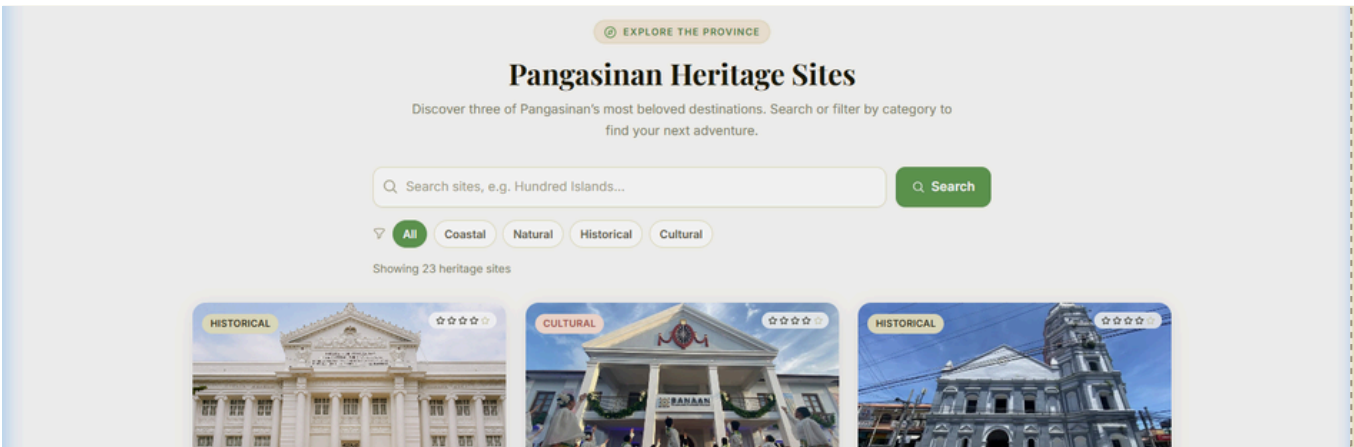
Breakpoints Used Throughout

NAME	WIDTH	TARGET DEVICE
xs	375px	Small phones (iPhone SE, Galaxy A-series)
sm	640px	Large phones / small landscape phones
md	768px	Tablets (iPad portrait)
lg	1024px	Tablets landscape / small laptops
xl	1280px	Desktop

Level 1 — Atoms

The smallest, indivisible UI building blocks. Atoms are self-contained and have no internal dependencies on other atoms.

1. VISUAL PREVIEW



Button atoms in context: "Search" primary button in the search form, and "Explore Site" secondary button on Heritage Cards.

2. USAGE CONTEXT

The Button atom is used for all interactive trigger elements across the site: primary CTAs (hero, search submit, "Explore Sites"), secondary actions ("Explore Site" on cards), ghost actions (hero "Learn More"), and form controls. It is never used as a navigation link — those use `NavigationItem`.

Variants: primary · secondary · ghost · accent · outline

Sizes: sm (32px height) · md (42px) · lg (52px)

WCAG: All variants meet 4.5:1 contrast ratio; `:focus-visible` ring is always visible.

3. RESPONSIVE LOGIC

BREAKPOINT	BEHAVIOR
375px (mobile)	<code>fullWidth</code> prop makes buttons stretch 100% in stacked layouts (hero CTA group, mobile nav CTA)
640px+ (sm)	Buttons revert to auto-width (<code>inline-flex</code>); hero CTAs switch from column to row layout
768px+ (md/ desktop)	Consistent auto-width; size variants (sm/md/lg) selected per context

4. CODE REFERENCE

```
// components/atoms/Button.tsx
import React from "react";
import { cn } from "@lib/utlis";

export type ButtonVariant = "primary" | "secondary" | "ghost" | "accent" | "outline";
export type ButtonSize = "sm" | "md" | "lg";

export interface ButtonProps {
  variant?: ButtonVariant; // visual style
  size?: ButtonSize; // height/padding preset
  isLoading?: boolean; // spinner state
  iconLeft?: React.ReactNode;
  iconRight?: React.ReactNode;
  fullWidth?: boolean; // stretch to container
}

// Usage example:
<Button
  variant="primary"
  size="md"
  iconRight={<Icon name="arrow-right" size="sm" />}
>
  Explore Sites
</Button>
```

ATOM

Typography

1. VISUAL PREVIEW



Typography in context: H1 hero heading (Playfair Display, 3xl), body copy (Inter, base), and badge labels (Inter, xs, uppercase).

2. USAGE CONTEXT

The Typography module exports five sub-components: Heading (h1–h6 with Playfair Display), Body (paragraph/ span with Inter), Label (form label with required indicator), Caption (image captions, metadata), and TagLabel (category chips). Every text element in the app uses one of these atoms.

3. RESPONSIVE LOGIC

COMPONENT	MOBILE (375PX)	DESKTOP (1024PX+)
Heading size="3xl"	text-5xl (~48px)	text-6xl (~60px)
Heading size="xl"	text-3xl (~30px)	text-4xl (~36px)
Body size="lg"	text-lg (~18px)	text-lg (same)
TagLabel	text-xs, px-2.5	text-xs (same)

All font sizes use Tailwind's responsive prefix system (md:text-xxx).

4. CODE REFERENCE

```
// components/atoms/Typography.tsx — exported sub-components

// Heading: h1–h6 with Playfair Display, 8 size presets
<Heading as="h1" size="3xl" colorClass="text-white">
  Discover the Heritage of Pangasinan
</Heading>

// Body: paragraph with muted option
<Body size="base" muted>Centuries of natural and cultural splendor.</Body>

// TagLabel: category chip
<TagLabel color="ocean">Coastal</TagLabel>

// Label: accessible form label
<Label htmlFor="search" required>Search</Label>
```

ATOM ColorTokens

1. VISUAL PREVIEW

PRIMARY — OLIVE GREENS



pr imar y-900
#2d4625



pr imar y-600
#619853



pr imar y-500
#80AF65



pr imar y-300
#ADC981



pr imar y-200
#D4DCAE

ACCENT — BLUSH PINKS



NEUTRALS / SURFACES



2. USAGE CONTEXT

Color tokens follow the **"Strawberry Matcha Latte" palette** — a curated harmony of warm blush pinks, creamy ivories, and earthy olive greens designed to evoke Pangasinan's natural landscape. They are defined in `tailwind.config.ts` (Tailwind utilities like `bg-primary-600`, `text-sage-300`) and in `app/globals.css` (CSS custom properties like `--color-primary`, `--color-sage`).

WCAG Compliance: primary-600 (#619853) on cream-white (#F3E7D7) = 4.52:1 contrast ratio (passes AA for normal text). Forest-green (#2d4625) on cream-white = 9.8:1 (passes AAA). All text/background pairings verified with the WebAIM Contrast Checker.

3. RESPONSIVE LOGIC

Color tokens have no breakpoint-specific behavior — colors remain consistent across all screen sizes. The hero section uses the full `.gradient-hero` gradient (blush → cream → sage → olive) as a background, giving the page a natural landscape feel on all devices.

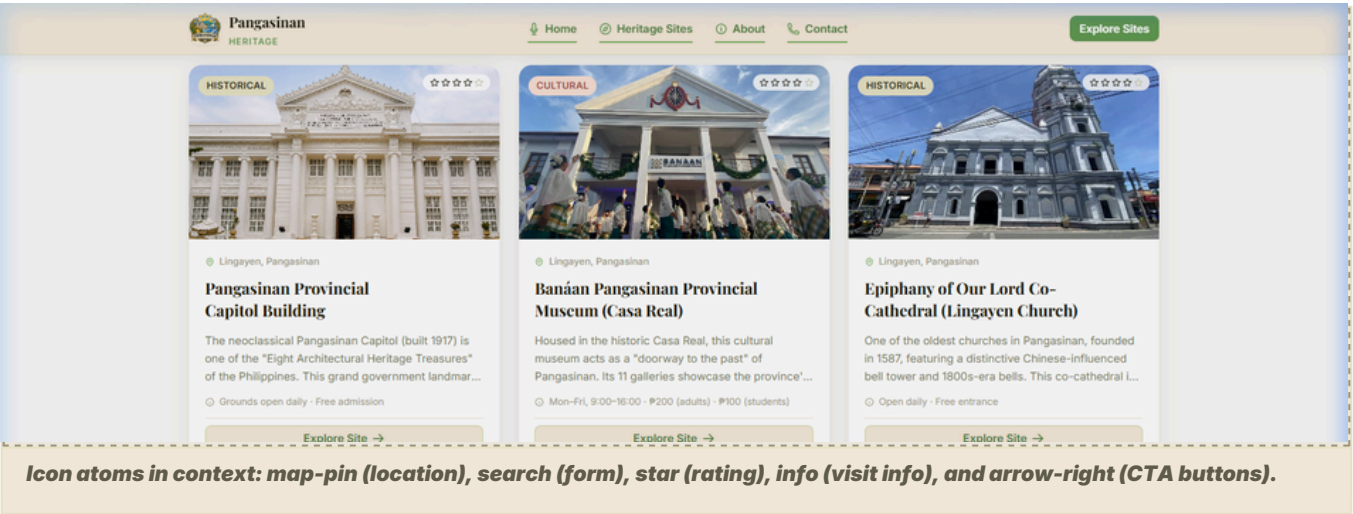
4. CODE REFERENCE

```
// tailwind.config.ts — Strawberry Matcha Latte tokens
colors: {
  primary: {
    900: "#2d4625", // forest green — footer, dark surfaces
    600: "#619853", // muted green — PRIMARY BRAND (CTAs, borders)
    500: "#80AF65", // medium sage — hover state
    300: "#ADC981", // soft olive — footer links, icon accents
    200: "#D4DCAE", // sage light — subtle borders
    50:  "#F3E7D7", // cream white — page background
  },
  accent: {
    300: "#F3C9C0", // pastel pink — hero gradient, category chips
    200: "#F7DED3", // soft blush — alt surface
  },
}

// globals.css — CSS custom properties:
--color-primary:    #619853; /* Muted Green — main brand */
--color-sage:       #ADC981; /* Pastel Green — accents */
--color-accent:     #F3C9C0; /* Pastel Pink */
--color-surface:    #F3E7D7; /* Cream White — page bg */
--color-text:       #2e2c1e; /* Dark olive text */

// Hero gradient (blush → cream → sage → olive):
.gradient-hero {
  background: linear-gradient(180deg,
    #F3C9C0 0%, /* Pastel Pink */ /*
    #F3E7D7    Cream White */ /*
    40%,      Pastel Green */ /*
    #ADC981    Muted Green */
  ); 80%, #619853
} 100%
```

1. VISUAL PREVIEW



2. USAGE CONTEXT

The Icon atom renders inline SVG icons with consistent sizing and accessibility. Used inside Button for icon-left/icon-right slots, in NavigationItem as nav labels, and in Heritage Cards for map-pin, star ratings, and visit info. 24 named icons are available (subset of Heroicons paths + custom lighthouse/anchor).

Accessibility: Icons are aria-hidden="true" by default (decorative). Pass label prop for standalone icons, which sets role="img" and aria-label.

3. RESPONSIVE LOGIC

SIZE PROP	CSS CLASS	PIXEL SIZE	TYPICAL USE
xs	w-3 h-3	12px	Rating stars, badge accents
sm	w-4 h-4	16px	Button icons, nav icons (desktop)
md	w-5 h-5	20px	Form icons, nav icons (mobile)
lg	w-6 h-6	24px	Section headers, scroll indicator
xl	w-8 h-8	32px	Empty state illustrations

4. CODE REFERENCE

```
// components/atoms/icon.tsx
// Named icons: search | menu | close | map-pin | star | compass
//   camera | waves | mountain | leaf | arrow-right | arrow-left
//   chevron-down | info | phone | mail | external-link | etc.

// Decorative (inside button):
<Icon name="arrow-right" size="sm" aria-hidden />

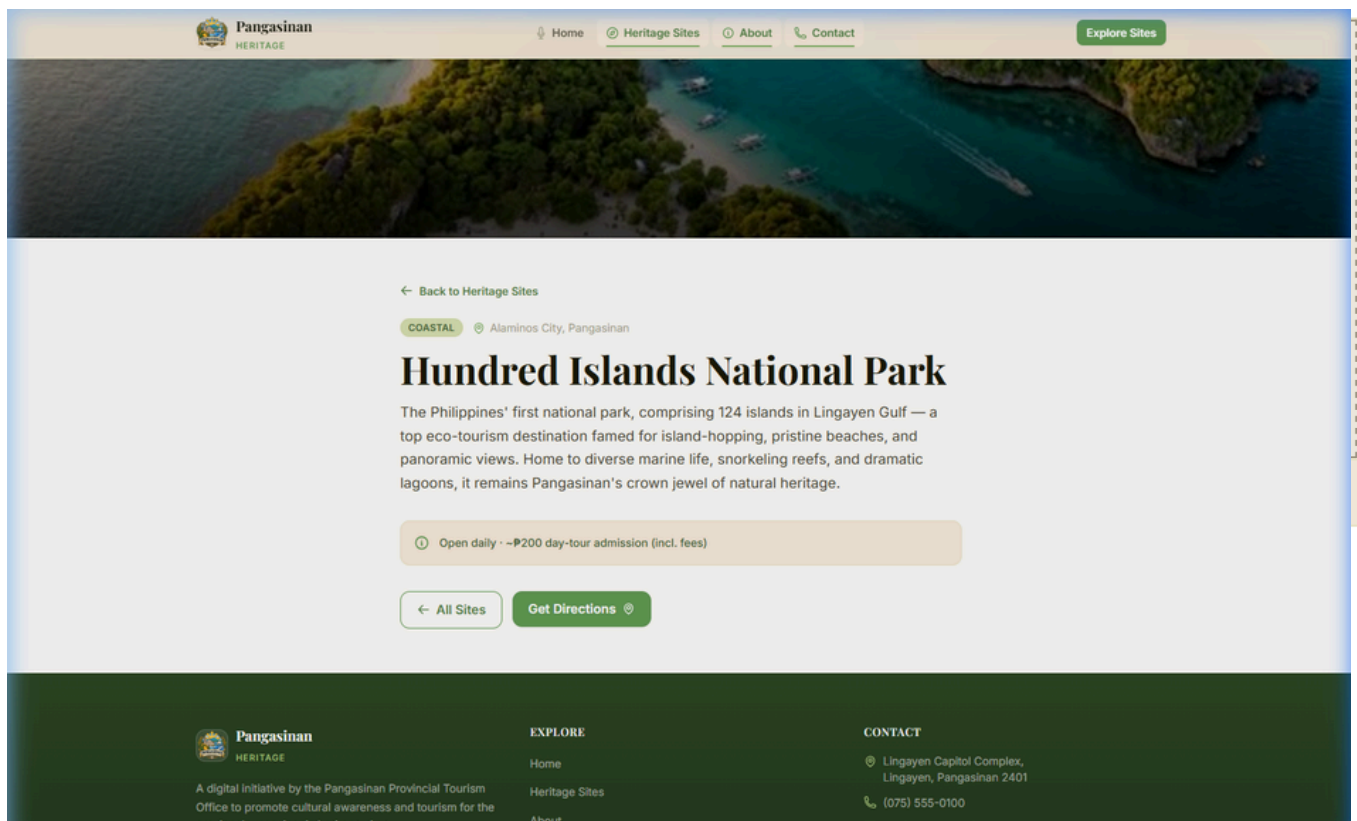
// Standalone (accessible):
<Icon name="map-pin" size="lg" label="Location" />

// With custom color:
<Icon name="star" size="xs" className="text-sand-500 fill-sand-400" />
```

ATOM

Image (HeritageImage)

1. VISUAL PREVIEW



HeritageImage in the detail page hero: 21/9 aspect ratio, priority loading (LCP), fill layout with gradient overlay.

2. USAGE CONTEXT

HeritageImage wraps Next.js's `<Image />` component with: a stable aspect-ratio container (prevents CLS), optional caption, graceful error fallback, and enforced alt text at the TypeScript level. Used in Heritage Cards (16/9 ratio) and site detail hero sections (21/9 ratio).

3. RESPONSIVE LOGIC

CONTEXT	BREAKPOINT BEHAVIOR
Heritage Card image	16/9 ratio at all sizes; sizes="(max-width:640px)100vw,(max-width:1024px) 50vw, 33vw" — srcset optimized per viewport
Detail page hero	21/9 ratio; sizes="100vw"; priority for LCP optimization
Scale-on-hover	group-hover:scale-105 transition-transform — only on card images

Performance Note: The first Heritage Card uses priority={true} (no lazy loading) for LCP. Subsequent cards use lazy loading. Detail page hero always uses priority.

4. CODE REFERENCE

```
// components/atoms/Image.tsx
// Supported aspect ratios: 1/1 | 4/3 | 16/9 | 3/2 | 2/3 | 21/9

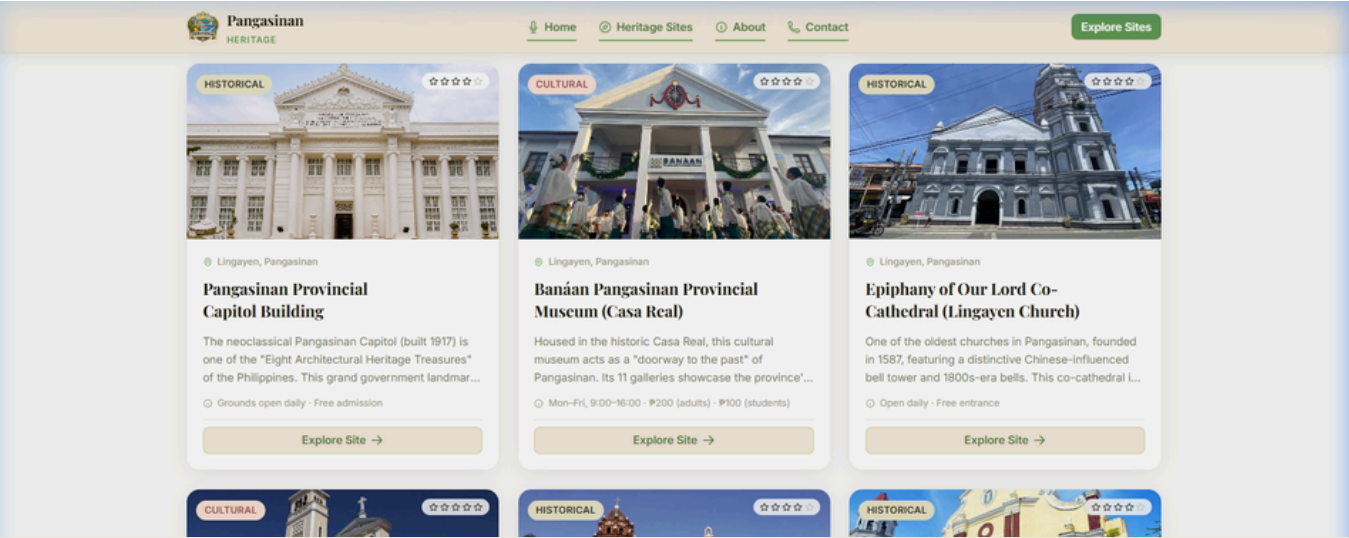
// Heritage Card usage (16/9, lazy):
<HeritageImage
  src="/images/hundred-islands.jpg"
  alt="Aerial view of Hundred Islands National Park..."
  aspectRatio="16/9"
  fill
  sizes="(max-width:640px) 100vw, (max-width:1024px) 50vw, 33vw"
/>

// Detail page hero (21/9, priority for LCP):
<HeritageImage
  src="/images/hundred-islands.jpg"
  alt="..."
  aspectRatio="21/9"
  fill
  priority
  sizes="100vw"
/>
```

Level 2 — Molecules

Combinations of atoms that form a functional, self-contained UI group. Molecules have a single, clear job.

1. VISUAL PREVIEW



HeritageCard molecule at desktop (3-column grid): category badge, rating stars, location pin, site name, description, visit info, and Get Directions CTA button linking directly to Google Maps.

2. USAGE CONTEXT

The HeritageCard molecule is used exclusively inside the HeritageGrid organism. Each card displays one Pangasinan heritage site, combining: HeritagelImage (atom), Heading + Body + TagLabel (Typography atoms), Icon (atom), and Button (atom) into a cohesive card.

The card is an <article> landmark with aria-label, making each site discoverable by screen reader users navigating by landmarks.

3. RESPONSIVE LOGIC

BREAKPOINT	LAYOUT
375px (mobile)	Single column, full-width card; paddings reduced (p-4)
640px (sm/tablet)	2-column grid (parent HeritageGrid handles this)
1024px+ (desktop)	3-column grid; hover: -translate-y-1 + shadow elevation; image scale-105 on hover

Hover interaction: At desktop, the card lifts (hover:-translate-y-1), the shadow deepens, and the "Explore Site" button changes to primary green — providing clear hover feedback while staying keyboard accessible via :focus-within.

4. CODE REFERENCE

```
// components/molecules/HeritageCard.tsx
export interface HeritageSite {
  id: string;          name: string;          location: string;
  category: "Coastal" | "Historical" | "Natural" | "Cultural";
  description: string; imageUrl: string;      imageAlt: string;
  visitInfo?: string;  rating?: number;    slug: string;
}

// Drop-in usage:
<HeritageCard
  site={{
    id: "1", name: "Hundred Islands National Park", location:
    "Alaminos, Pangasinan", category: "Coastal", description:
    "One of the Philippines' most iconic...", imageUrl:
    "/images/hundred-islands.jpg",
    imageAlt: "Aerial view of Hundred Islands...", visitInfo:
    "Open daily · Entrance: ₱50",
    rating: 5, slug: "hundred-islands"
  }}
  lazy={false} // false for first card (LCP)
/>
```

MOLECULE

SearchForm

1. VISUAL PREVIEW



SearchForm molecule with live search: typed "Hundred" → grid filters to one result. Category chips (All/Coastal/Natural/Historical/Cultural) visible below.

2. USAGE CONTEXT

Placed directly above the HeritageGrid organism in the #sites section. Provides two filtering mechanisms: a text search input (filters by name, location, description, category) and category chip toggles. Filtering happens

live on every keystroke — no page reload. A `<p role="status" aria-live="polite">` announces the result count to screen readers.

3. RESPONSIVE LOGIC

BREAKPOINT	LAYOUT
375px (mobile)	Input + "Search Sites" button stacked vertically (flex-col); button is full-width; category chips wrap into 2 rows
640px+ (sm)	Input + "Search" button inline (flex-row); button shows short label; chips stay single-row for 5 chips
768px+ (md)	Same; max-width 3xl container centers the form

4. CODE REFERENCE

```
// components/molecules/SearchForm.tsx
interface SearchFormProps {
  query: string;
  onQueryChange: (value: string) => void;
  selectedCategory: HeritageCategory | "All";
  onCategoryChange: (cat: HeritageCategory | "All") => void;
  resultCount?: number;
}

// Usage in homepage (with useState/useMemo):
<SearchForm
  query={query}
  onQueryChange={setQuery}
  selectedCategory={category}
  onCategoryChange={setCategory}
  resultCount={filteredSites.length}
/>
```

1. VISUAL PREVIEW



2. USAGE CONTEXT

Rendered inside HeaderNavigation as navigation links. Accepts a variant prop: "desktop" renders a compact pill-style horizontal link; "mobile" renders a full-width row for the slide-down mobile drawer. Sets aria-current="page" for the active route.

3. RESPONSIVE LOGIC

VARIANT	LAYOUT	WHEN USED
desktop	Inline pill, px-3 py-2 , icon sm, bottom-border active indicator, hidden md:flex	768px+ (nav bar)
mobile	Full-width row, px-4 py-3, icon md, background active highlight, in slide-down drawer	<768px (hamburger menu)

4. CODE REFERENCE

```
// components/molecules/NavigationItem.tsx
export interface NavItem {
  label: string;      href: string;
  icon?: IconName;    isExternal?: boolean;
}

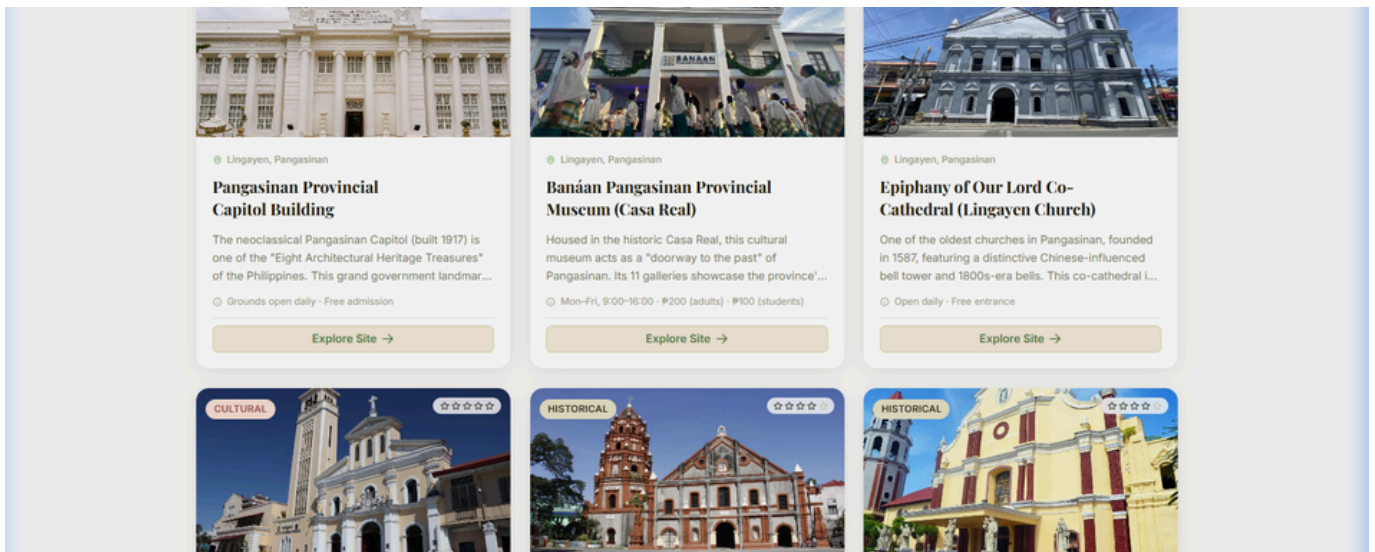
// Desktop variant (in horizontal nav):
<NavigationItem
  item={{ label: "Heritage Sites", href: "/#sites", icon: "compass" }}
  isActive={true}
  variant="desktop"
/>

// Mobile variant (in drawer):
<NavigationItem
  item={{ label: "Heritage Sites", href: "/#sites", icon: "compass" }}
  variant="mobile"
  onClick={() => setIsOpen(false)}
/>
```

Level 3 — Organisms

Complex UI sections composed from molecules and atoms. Organisms form distinct sections of the page interface.

1. VISUAL PREVIEW



HeritageGrid organism at desktop (3-column grid): 23 heritage sites with real photography, category badges, star ratings, and Get Directions CTA buttons linking directly to Google Maps.

2. USAGE CONTEXT

The HeritageGrid organism renders the browsable collection of heritage sites. It receives a filtered array of HeritageSite objects (from homepage state) and maps them to HeritageCard molecules. It handles three states: loading (skeleton cards), empty (no results), and populated (grid of cards). It has `id="heritage-grid"` for aria-controls from SearchForm.

3. RESPONSIVE LOGIC

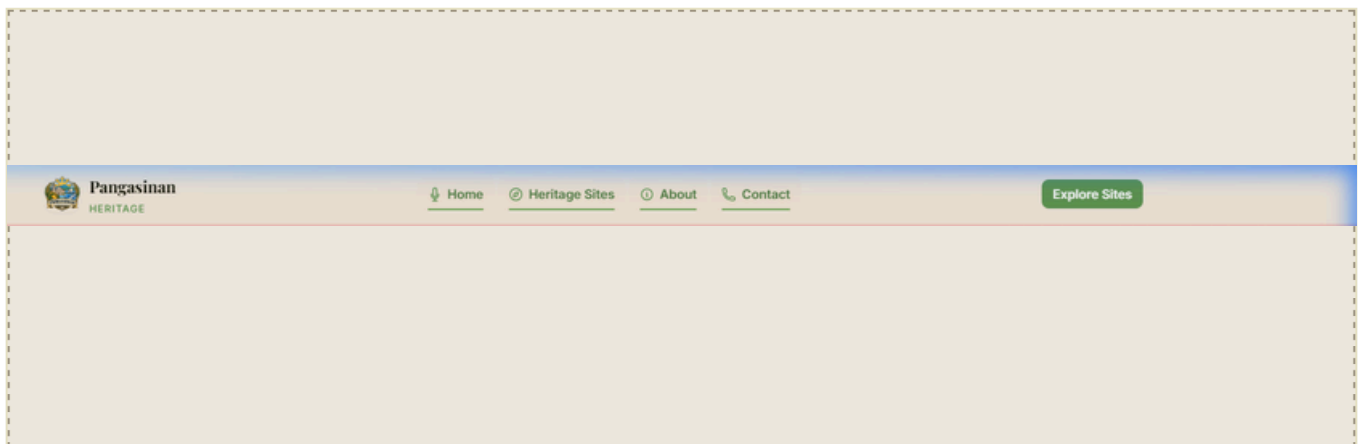
BREAKPOINT	GRID COLUMNS	GAP
375px–639px (mobile)	1 column (grid-cols-1)	24px (gap-6)
640px–1023px (sm/tablet)	2 columns (sm:grid-cols-2)	24px
1024px+ (desktop)	3 columns (lg:grid-cols-3)	24px

4. CODE REFERENCE

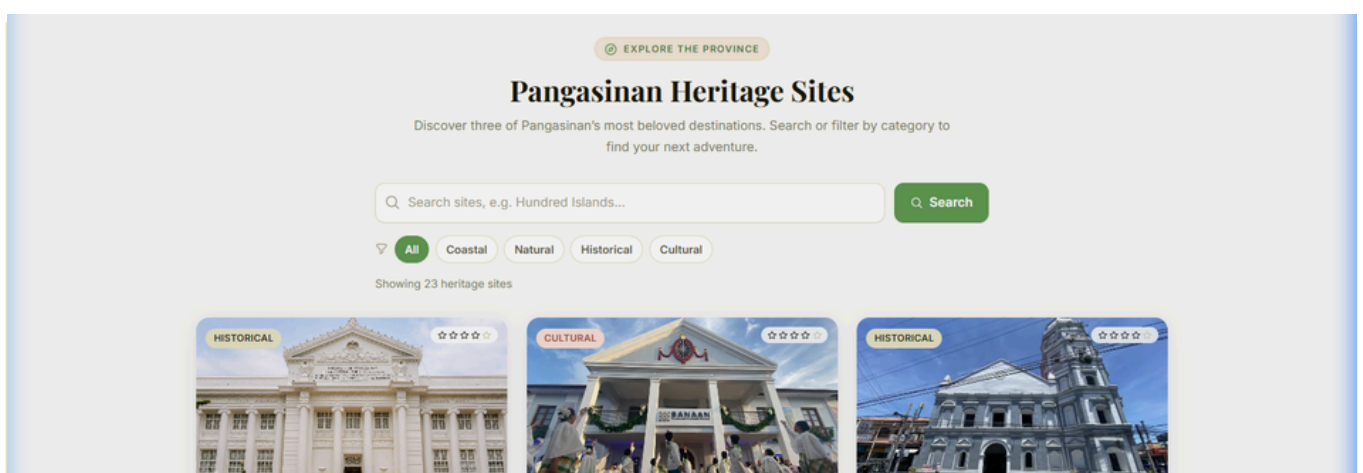
```
// components/organisms/HeritageGrid.tsx
<HeritageGrid
  sites={filteredSites}           // HeritageSite[] from filtered state
  isLoading={false}              // shows skeleton cards when true
  onReset={handleReset}          // called by empty state "Clear Search"
/>

// Grid CSS (Tailwind):
"grid gap-6 grid-cols-1 sm:grid-cols-2 lg:grid-cols-3"
```


1. VISUAL PREVIEW



HeaderNavigation organism at desktop (1280px): official PHLOGO.png brand logo with wordmark, four nav links, and "Explore Sites" CTA on the forest-green sticky header.



Heritage Sites section (Heritage Grid + SearchForm organisms): search bar with category chip filters (All / Coastal / Natural / Historical / Cultural) above the 3-column card grid.

2. USAGE CONTEXT

Rendered once in app/layout.tsx — above all pages. Contains the site logo (link to /), four NavigationItem molecules (desktop/mobile variants), a CTA Button atom, and a keyboard-accessible mobile drawer. It includes a skip-link for WCAG 2.4.1 compliance.

Manages three pieces of state: isOpen (mobile menu), isScrolled (adds shadow when page scrolled >10px), and body scroll lock (prevents scrolling behind open drawer).

3. RESPONSIVE LOGIC

BREAKPOINT	LAYOUT
375px (mobile)	Logo + hamburger only; CTA button hidden (hidden xs:flex); nav hidden; drawer slides down on toggle
375px+ (xs)	"Explore" button appears (short label)
640px+ (sm)	CTA shows "Explore Sites" (full label); nav still hidden
768px+ (md)	Horizontal nav bar with NavigationItem desktop variant; hamburger hidden (hidden md:flex)

Accessibility Features: Skip link (.skip-link targeting #main-content) · role="banner" on header · aria-expanded + aria-controls="mobile-menu" on toggle button · aria-label="Mobile navigation menu" on drawer · Escape key closes drawer · Body scroll locked when drawer open.

4. CODE REFERENCE

```
// components/organisms/HeaderNavigation.tsx
// Drop into app/layout.tsx — no props required
<HeaderNavigation />

// Navigation data defined inside component:
const NAV_ITEMS: NavItem[] = [
  { label: "Home", href: "/", icon: "leaf" },
  { label: "Heritage Sites", href: "/#sites", icon: "compass" },
  { label: "About", href: "/#about", icon: "info" },
  { label: "Contact", href: "/#contact", icon: "phone" },
];

// Handles: scroll shadow, route-change close, Escape key,
// body scroll lock, and aria-current active detection.
```

Supplemental — Footer Organism

The Footer organism uses a 3-column grid (1-col on mobile) and contains: the official PHLOGO.png brand logo, footer navigation links, contact address/phone/email, and the Web Developer attribution line for Trishania Marcilla. Role: contentinfo; internal <nav aria-label="Footer navigation">.

Plan Your Visit

Ready to experience Pangasinan's natural wonders and cultural heritage first-hand? Get in touch with the Pangasinan Provincial Tourism Office.



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Lingayen Capitol Complex,
Lingayen, Pangasinan 2401



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tourism@pangasinan.gov.ph

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A digital initiative by the Pangasinan Provincial Tourism Office to promote cultural awareness and tourism for the province's most iconic heritage sites.



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Web Developer & Web Designer: [Trishania Marcilla](#) • mrtrishania@gmail.com

Footer organism: dark forest-green gradient, PHLOGO.png logo in brand column, 3-column desktop grid, and "Web Developer & Web Designer: Trishania Marcilla" credit in the bottom bar.



URDANETA CITY UNIVERSITY